

XiangShan: An Open-Source High-Performance RISC-V Processor and Infrastructure for Architecture Research

The XiangShan Team

ISCA'25@Tokyo, Japan

June 21, 2025

Together for a shared future

- RVSE'25 @ Paris, France
 - ASPLOS'25 @ Rotterdam, The Netherlands
 - HPCA'25 @ Las Vegas, USA
 - MICRO'24 @ Austin, USA
 - ASPLOS'24 @ San Diego, USA
 - HPCA'24 @ Edinburgh, Scotland
 - MICRO'23 @ Toronto, Canada
 - ASPLOS'23 @ Vancouver, Canada
-
- English Biweekly including latest progress and performance data
 - Our official document is continuously updating
 - Close connections with open-source community
 - Feel free to contact us through email or file issues on GitHub!
 - all@xiangshan.cc
 - <https://github.com/OpenXiangShan/XiangShan>



XiangShan Home page



XiangShan Document

What we will cover in this tutorial

- **Introduce of XiangShan (14:00 – 14:30, 30 minutes)**
- **Introduction of XS-Gem5 (14:30 – 15:00, 30 minutes)**
- **CPU Microarchitecture (15:00 – 16:00, 60 minutes)**
 - Design and implementation – How to implement novel ideas on XiangShan
 - Frontend: branch prediction and instruction fetch
 - Backend: out-of-order scheduler, execution units
 - Load/Store Unit: LSQ, pipelines, TLBs, data caches
 - L2/L3 caches and prefetchers
- **Development workflows (16:30 – 18:00, 90 minutes)**
 - Introduction of their usages – How to develop on XiangShan with MinJie
 - Simulation and Debugging
 - Research Demo



ISCA 2025

Part I

The Era of Open-Source Chip





A chip design that changes everything

- 10 Breakthrough Technologies 2023

Ever wonder how your smartphone connects to your Bluetooth speaker, given they were made by different companies? Well, Bluetooth is an open standard, meaning its design specifications, such as the required frequency and its data encoding protocols, are publicly available. Software and hardware based on open standards—Ethernet, Wi-Fi, PDF—have become household names.

Now an open standard known as RISC-V (pronounced “risk five”) could change how companies create computer chips.

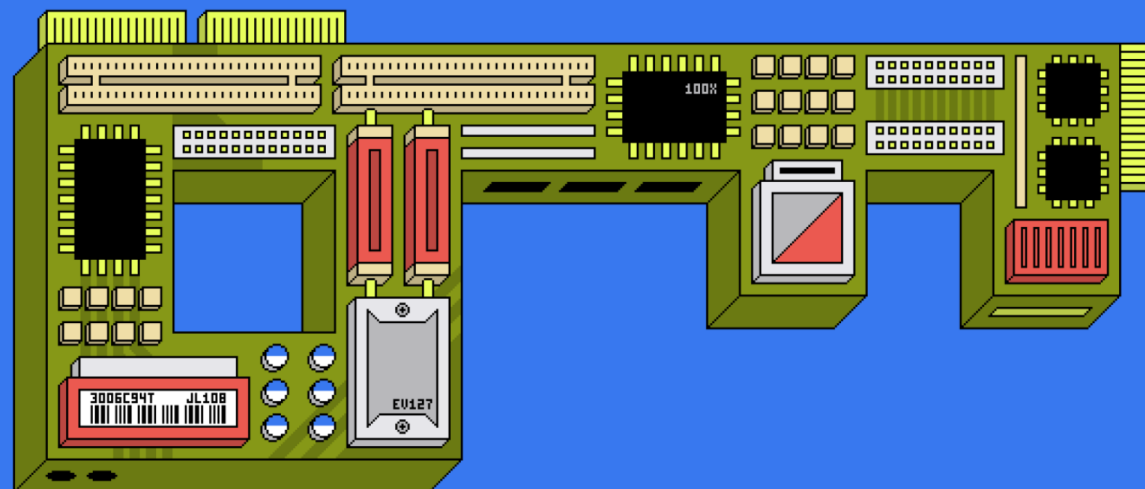
--- MIT Technology Review

A chip design that changes everything: 10 Breakthrough Technologies 2023

Computer chip designs are expensive and hard to license. That's all about to change thanks to the popular open standard known as RISC-V.

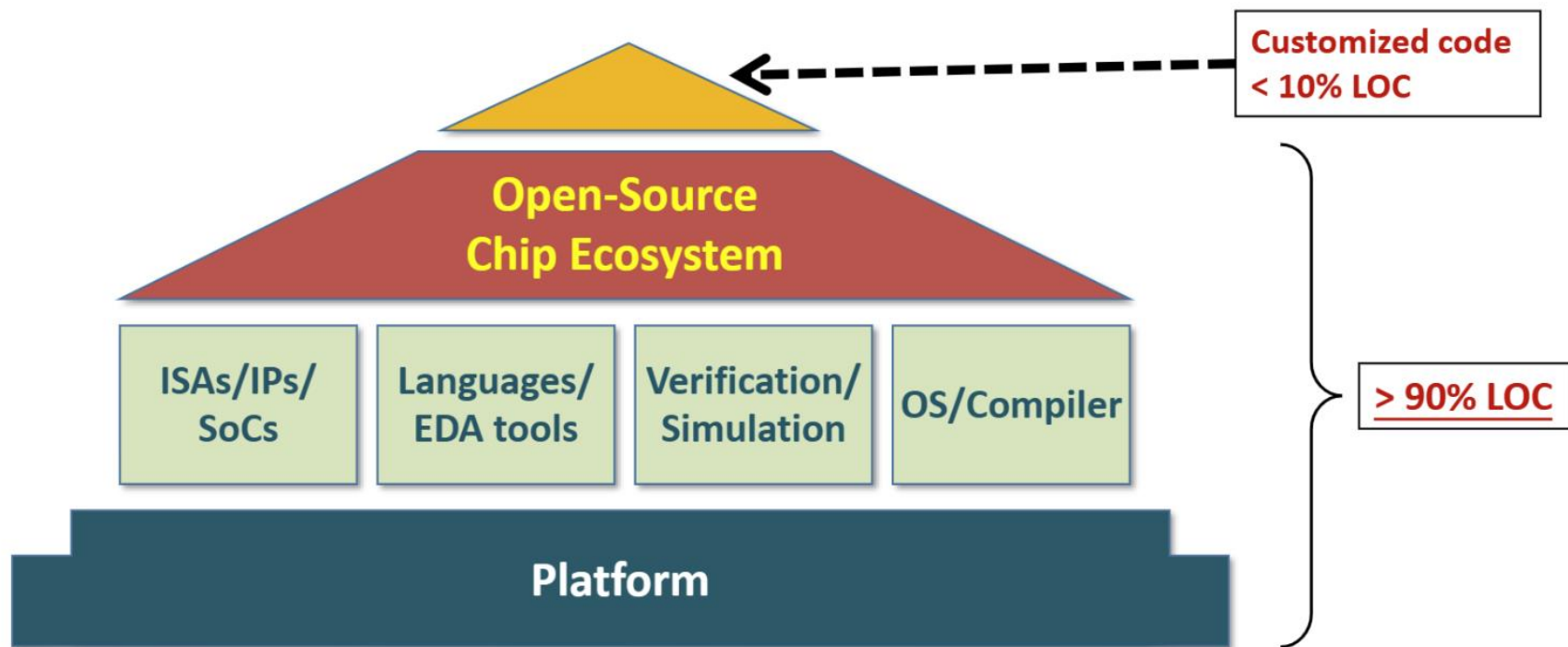
By Sophia Chen

January 9, 2023



🏔️ Open-Source Chip Ecosystem (OSCE)

- Goal: mirror the success of the open-source software ecosystem



To Lower the barrier of chip development

By saving the cost of IPs, EDA tools and engineers in chip design



Three Levels of Open-Source Chip

L1: OPEN ISA

L2: OPEN Design/Implementation

L3: OPEN Framework/Tools

Open Framework

Open Flow

Open Tools

microarch.

coding

EDA Tools

ISA Spec.

Docs

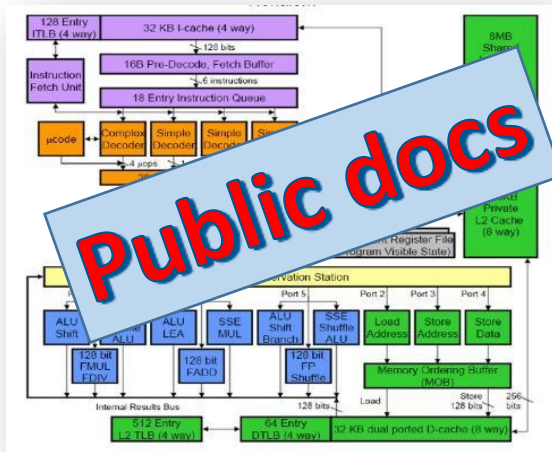
RTL

Layout

Open & Free

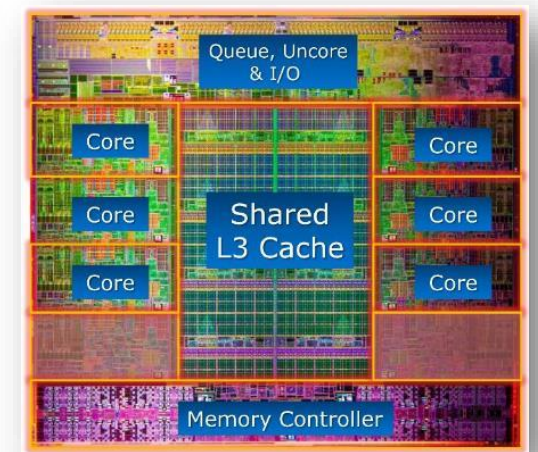
Public docs

Open-source



```
component DebugCoreTop is
  port (
    -- Trigger and Data
    cu_Clk      : in  std_logic_vector(10 downto 0);
    cu0_Trig    : in  t_trig_0;
    cu1_Trig    : in  t_trig_1;
    cu2_Trig    : in  t_trig_2;
    cu0_Data    : in  std_logic_vector(31 downto 0);
    cu1_Data    : in  std_logic_vector(31 downto 0);
    cu2_Data    : in  std_logic_vector(31 downto 0);

    gt_Clk      : in  std_logic := '0';
    gt_RefClk_n : in  std_logic := '0';
    gt_RX_p     : in  std_logic_vector(2 downto 0) := (others => '0');
    gt_RX_n     : in  std_logic_vector(2 downto 0) := (others => '0');
    gt_TX_p     : out std_logic_vector(2 downto 0);
    gt_TX_n     : out std_logic_vector(2 downto 0);
  );
end component;
```





Three levels of OSCE

L1: OPEN ISA

L2: OPEN Design & Implementation

L3: OPEN Tools & Infrastructure

3

Open Tools & Infrastructure

Microarch.

Engineering

EDA Tools

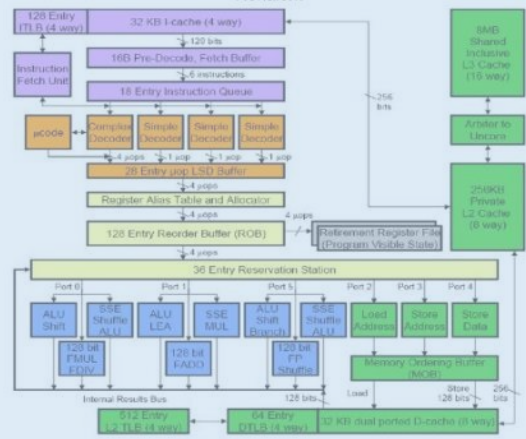
ISA Spec.



1

Open ISA

Docs



2

Open Design & Implt

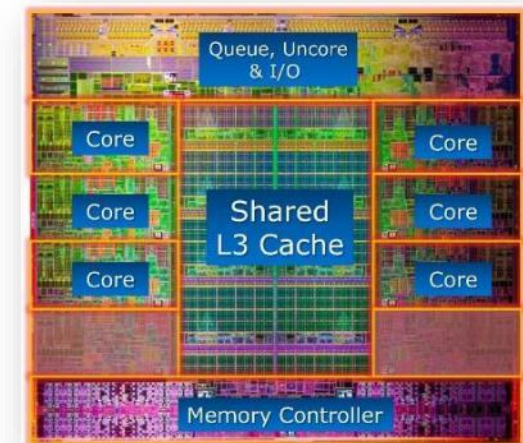
RTL codes

```
component DebugCoreTop is
  port (
    -- Trigger and Data
    cu_Clk      : in  std_logic_vector(2 downto 0) := (others => '0');
    cu0_Trig    : in  t_trig_0 := (others => (others => '0'));
    cu1_Trig    : in  t_trig_1 := (others => (others => '0'));
    cu2_Trig    : in  t_trig_2 := (others => (others => '0'));
    cu0_Data    : in  t_data_0 := (others => (others => '0'));
    cu1_Data    : in  t_data_1 := (others => (others => '0'));
    cu2_Data    : in  t_data_2 := (others => (others => '0'));

    -- Downstream I2C
    SCL         : in  std_logic := '0';
    SDA         : inout std_logic := '0';

    -- Upstream
    gt_RefClk_p : in  std_logic := '0';
    gt_RefClk_n : in  std_logic := '0';
    gt_RX_p     : in  std_logic_vector(2 downto 0) := (others => '0');
    gt_RX_n     : in  std_logic_vector(2 downto 0) := (others => '0');
    gt_TX_p     : out std_logic_vector(2 downto 0);
    gt_TX_n     : out std_logic_vector(2 downto 0)
  );
end component;
```

Layout





Why Open-Source High-perf. RISC-V Processor XiangShan?

- **Why RISC-V:** Free and open ISA
- **Why high-perf :** Most RISC-V processors are for IoT/edge, but both academic and industrial community need high-performance RISC-V processors
- **Why open-source:** An open and innovative hardware platform
- Build a leading platform with end-to-end agile development flows and tools



Linux

V.S.

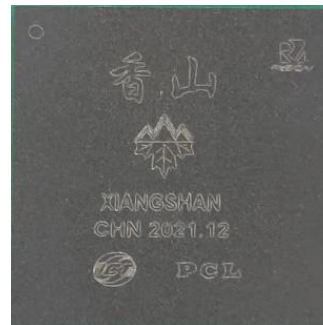


XIANGSHAN

**Envision :
“Hardware Version of Linux”**

Part II

XiangShan: Open-Source High Performance RISC-V Processor





XiangShan: Open-Source High Performance Processors



L1: OPEN ISA

L2: OPEN Design/Implementation

L3: OPEN Framework/Tools



Fragrant Hill in Beijing

XiangShan Public

Edit Pins Unwatch 96 Fork 748 Starred 6.2k

master 178 Branches 5 Tags

Go to file Add file Code

About

Open-source high-performance RISC-V processor

xiangshan.cc

chisel risc-v microarchitecture

Readme View license Activity Custom properties 6.2k stars 96 watching 748 forks Report repository

Siudya and cz4e feat(dfx): reuse dcache data sram read data register as mbist ... ebe07d6 · 2 hours ago 10,757 Commits	
.github	Revert "feat(AIA): integrate ChiselAIA (#4378)" (#4429) 3 days ago
coupledL2 @ 851b04a	submodule(CoupledL2): bump CoupledL2 (#4384) last week
debug	bump difftest & mkdir for wave/perf for local-ci script's run-... 2 years ago
difftest @ 62e4540	feat(Difftest): add multi-core vector load check (#4361) 2 weeks ago
fudian @ e1bd469	chore: bump chisel 6.0.0 (#2654) last year
huancun @ 90aaf59	feat(TopDown): add TopDown PMU Events (#4122) 2 months ago
images	docs(readme): update kunminghu architecture graph (#3910) 4 months ago
macros/src/main/scala	NewCSR: fix unprivileged CSRs and permission check 8 months ago

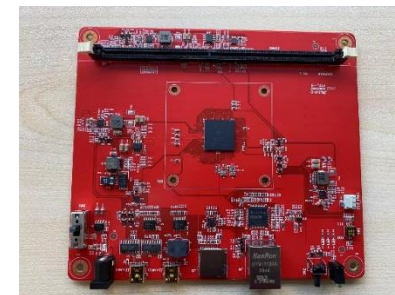
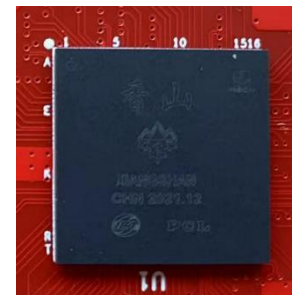
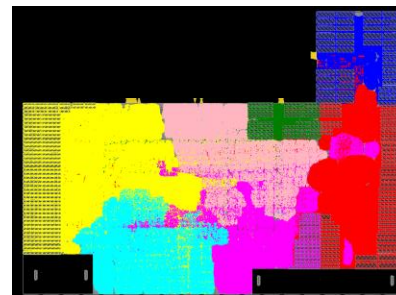
> 6.2K stars, > 740 forks on GitHub



XiangShan: Open-Source High Performance Processors

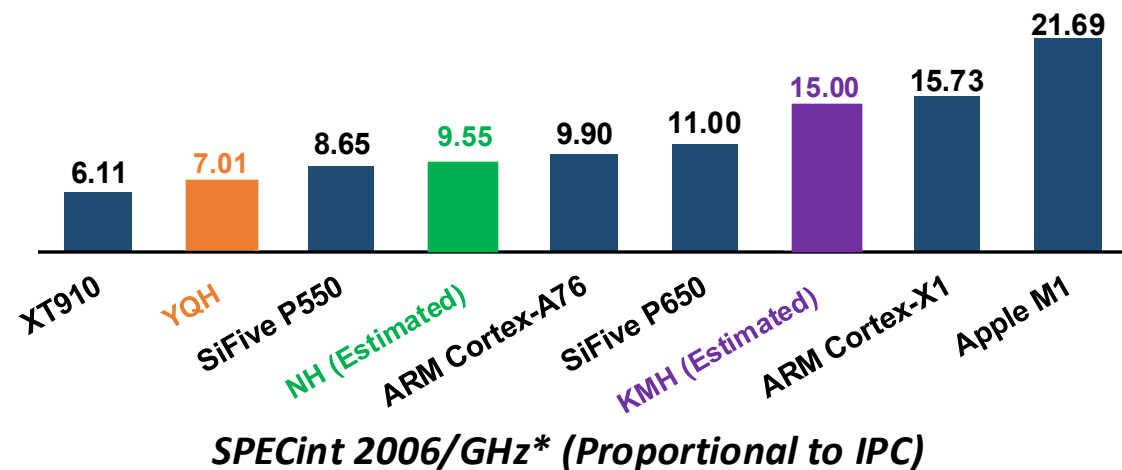
- **1st generation: Yanqihu (YQH)**

- RV64GC, single-core, superscalar OoO
- 28nm tape-out, 1.3GHz, July 2021
- SPEC CPU2006 7.01@1GHz, DDR4-1600



- **2nd generation: Nanhu (NH)**

- RV64GCBK, dual-core, superscalar OoO
- 14nm GDSII delivery, 2GHz, 2023 Q3
- Estimated** SPECint 2006 19.10@2GHz



- **3rd generation: Kunminghu (KMH)**

- RV64GCBKHV, quad-core, superscalar OoO
- Advanced-node, 3GHz, 1.5x IPC of NH
- Close collaboration with industrial partners

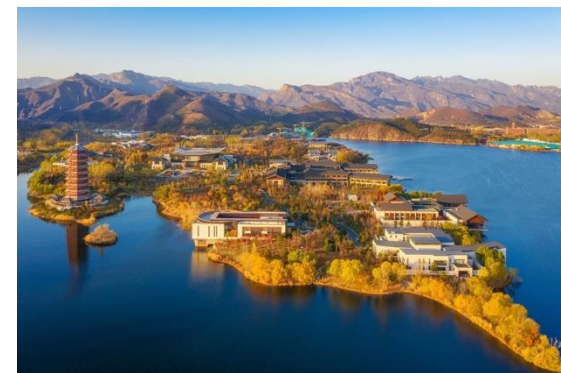


* Source: XT910@ISCA'20, SiFive, AnandTech

** Updated January 5, 2023

XiangShan Gen 1: Yanqihu

- **Test chip developed almost entirely by students**
 - RV64GC, 11-stage, superscalar, out-of-order
 - 5.3 CoreMark/MHz (gcc-9.3.0 -O2)
- **Tape-out:** single XiangShan core with 1MB L2 Cache
 - 28nm, 1.3GHz, July 2021



Yanqi Lake in Beijing

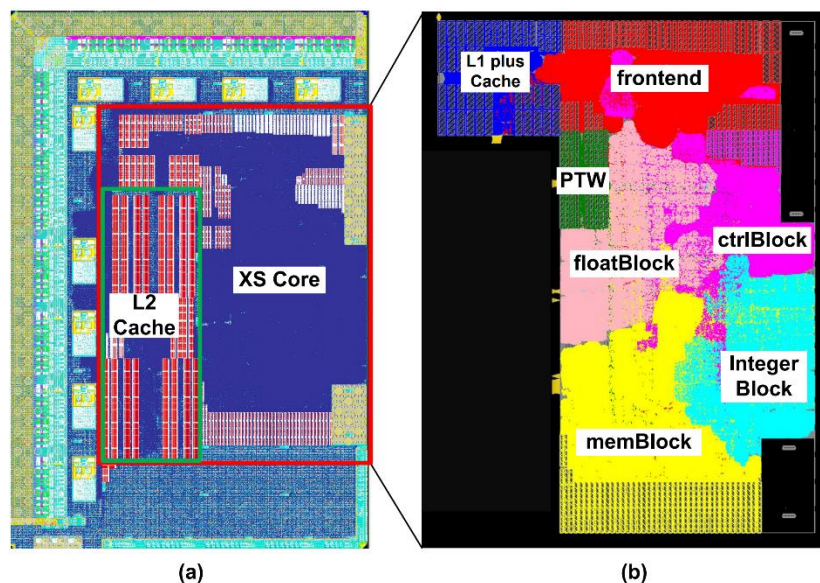


Figure. Layout of (a) the entire chip; (b) the core

Tape-out information for the processor core

Process Node	28nm
Die Size	8.6 mm ²
Std Cell	5.05M, 4.27 mm ²
Mem	261, 1.7mm ²
Density	66%
Cell	ULVT 1.04%, LVT 19.32%, SVT 25.19%, HVT 53.67%
Estimated Power	5W
Frequency	1.3GHz, TT85C

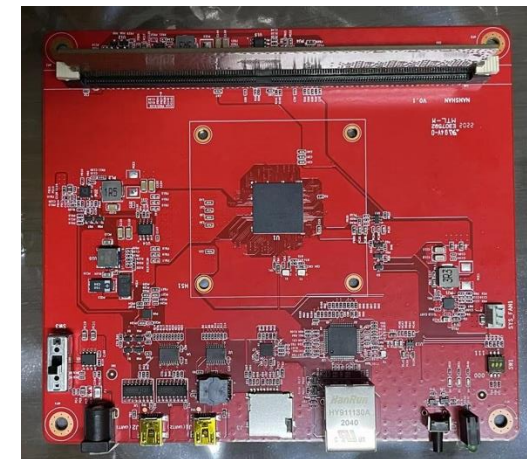
Real Chip of XiangShan Gen 1 Yanqihu

- The chip was back in January 2022
 - SoC: CPU, SPI Flash, UART, SD card, Ethernet, DIMM
 - Correctly running Debian with SD card and ethernet
- Performance: SPEC CPU2006 7.01@1GHz

SPECint 2006 @ 1GHz		SPECfp 2006 @ 1GHz	
400.perlbench	6.14	410.bwaves	9.28
401.bzip2	4.37	416.gamess	6.59
403.gcc	6.71	433.milc	8.41
429.mcf	6.83	434.zeusmp	7.65
445.gobmk	7.92	435.gromacs	4.99
456.hmmer	5.24	436.cactusADM	3.97
458.sjeng	6.85	437.leslie3d	6.93
462.libquantum	17.71	444.namd	8.00
464.h264ref	10.91	447.dealII	10.17
471.omnetpp	5.65	450.soplex	7.03
473.astar	5.16	453.povray	7.14
483.xalancbmk	7.35	454.Calculix	2.86
		459.GemsFDTD	8.35
		465.tonto	6.42
		470.lbm	10.39
		481.wrf	7.26
		482.sphinx3	9.07

SPECint 2006: 7.03@1GHz

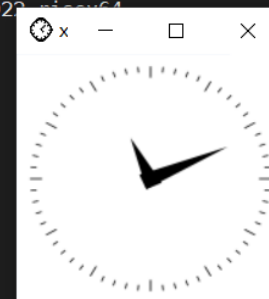
SPECfp 2006: 7.00@1GHz



```
wanghuizhe@open02:~$ ssh -X xs@172.28.2.246
xs@172.28.2.246's password:
Linux open02 4.20.0-44668-ge9c195ab0c63-dirty #109 Thu Feb 17 17:41:13 CST 2022; root@open02

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
You have no mail.
Last login: Thu Feb 17 11:10:31 2022 from 172.28.9.102
xs@open02:~$ xclock
Warning: locale not supported by C library, locale unchanged
```



**SSH into the Debian on XiangShan, and
run a GUI program via X11 forwarding**

XiangShan Gen 2: Nanhu

- **Target: 2GHz@14nm, SPEC CPU2006 ~20 marks**
- **New frontend:** decoupled BP and instruction fetch
- **Improved backend:** better scheduler, instruction fusions, move elimination, and more
- **New L2/L3 cache:** designed for high frequency and high performance with hybrid prefetchers
- **Verified under dual-core config (RV64GCBK),** more devices support (PCIe, USB, ...)

Setup an open and standardized development workflow



A lake in Jiaying, Zhejiang, China

☐		10 Open	✓	2,322 Closed
☐		Bump difftest for palladium simulation ✓		
		#2662 opened 11 hours ago by klin02 • Approved		
☐		ICache: fix ICacheMainPipe bug about sfence ✓	bug	do not merge
		#2660 opened 3 days ago by ssszwic • Review required		
☐		ICache: change data SRAM partWayNum from 2 to 4 ✓		
		#2653 opened 5 days ago by ssszwic • Approved		
☐		ITTAGE meta width shrink ✓	do not merge	
		#2592 opened last month by eastonman • Review required		
☐		pf: fix negative stream ✓	bug	
		#2508 opened on Nov 27, 2023 by happy-lx • Review required		
☐		SQ: add sq merge ✓	do not merge	enhancement
		#2439 opened on Oct 29, 2023 by sfencevma • Review required		
☐		rm refillPipe ✓	enhancement	
		#2426 opened on Oct 25, 2023 by YukunXue • Approved		

Pull Request Snapshot

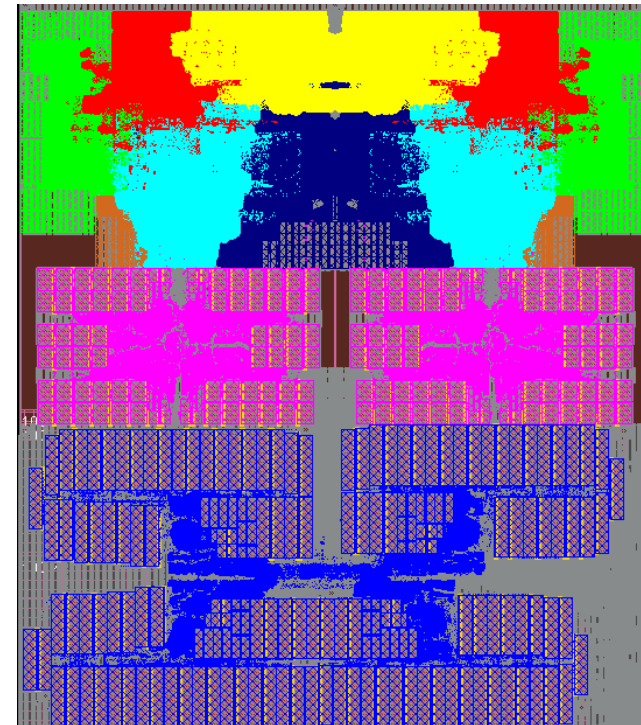


Estimated Performance of XiangShan Gen 2 Nanhu

- Estimated **SPECint 2006 19.10, SPECfp 2006 22.18@2GHz**
 - RTL simulation, DDR4-2400 under DRAMsim3
 - Compile with GCC 10.2.0, -O2, RV64GCB

SPECint 06	
400.perlbench	19.27
401.bzip2	11.36
403.gcc	21.97
429.mcf	20.53
445.gobmk	15.97
456.hmmer	19.22
458.sjeng	17.22
462.libquantum	36.99
464.h264ref	28.54
471.omnetpp	14.02
473.astar	14.19
483.xalancbmk	21.48
SPECint2006@2GHz	19.10

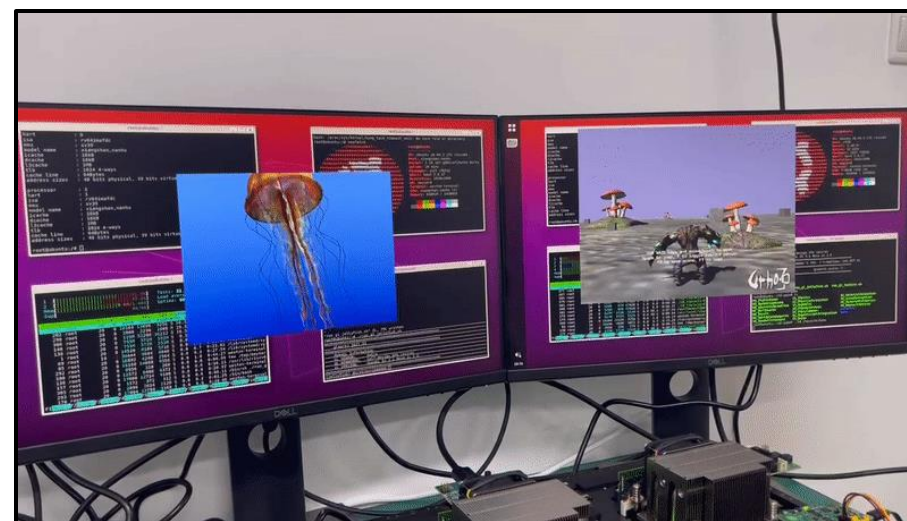
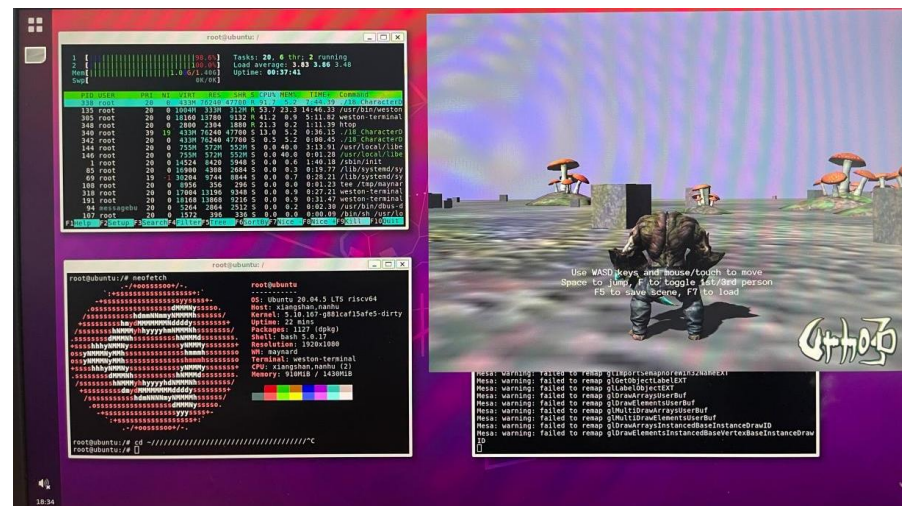
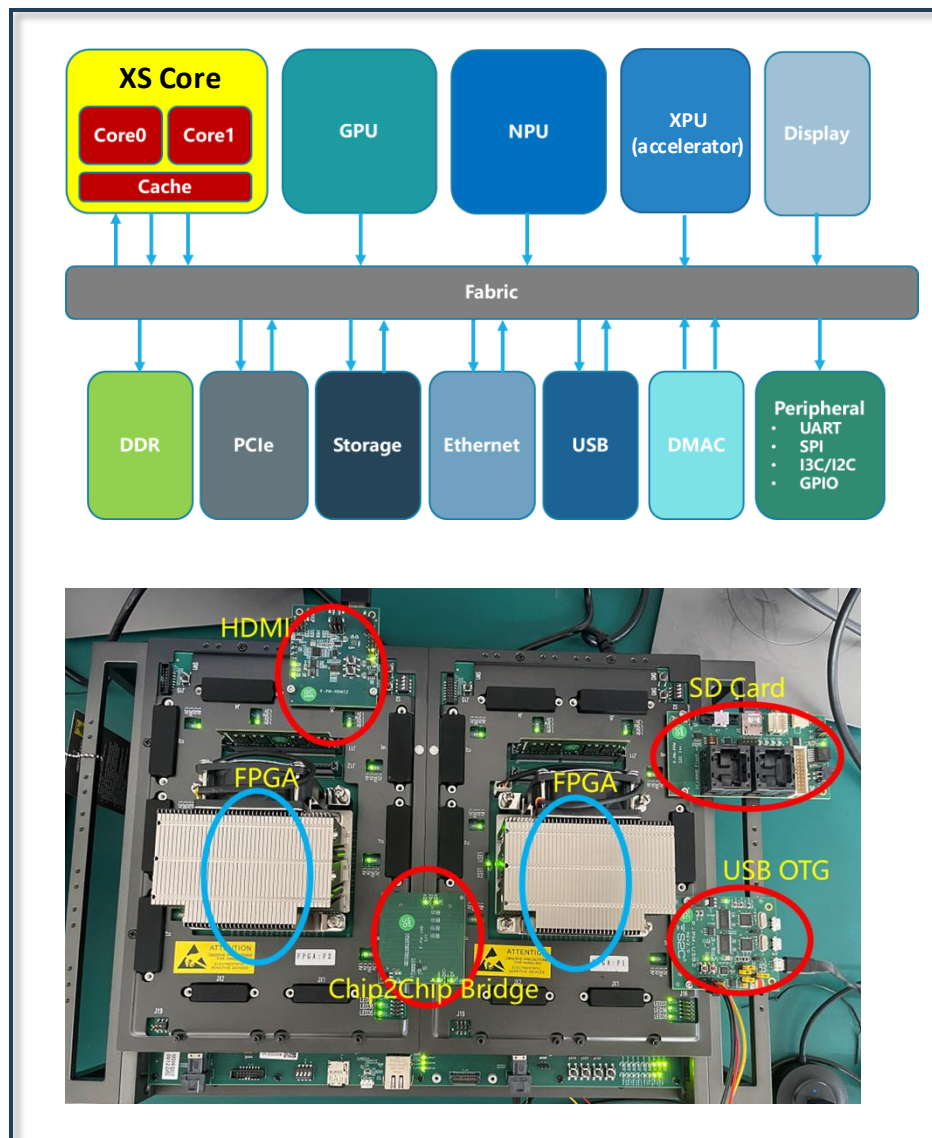
SPECfp 06	
410.bwaves	18.09
416.gamess	23.82
433.milc	18.33
434.zeusmp	28.19
435.gromacs	17.53
436.cactusADM	24.26
437.leslie3d	20.28
444.namd	23.83
447.dealII	33.50
450.soplex	25.61
453.povray	27.06
454.Calculix	9.18
459.GemsFDTD	24.66
465.tonto	17.68
470.lbm	32.04
481.wrf	19.73
482.sphinx3	28.38
SPECfp2006@2GHz	22.18



Feature	V1 YQH	V2 NH
ISA	RV64GC	RV64GCBK
Process Node	28nm	14nm
Core Count	1	2
Die Size	8.6 mm ²	22.13 mm ²
Std Cell Num/Area	5.05M, 4.27 mm ²	11.3M, 4.53 mm ²
Mem Num/Area	261, 1.7 mm ²	692, 8.93 mm ²
Density	66%	35%
Frequency	1.3GHz, TT 0.9V	2GHz, TT 0.9V



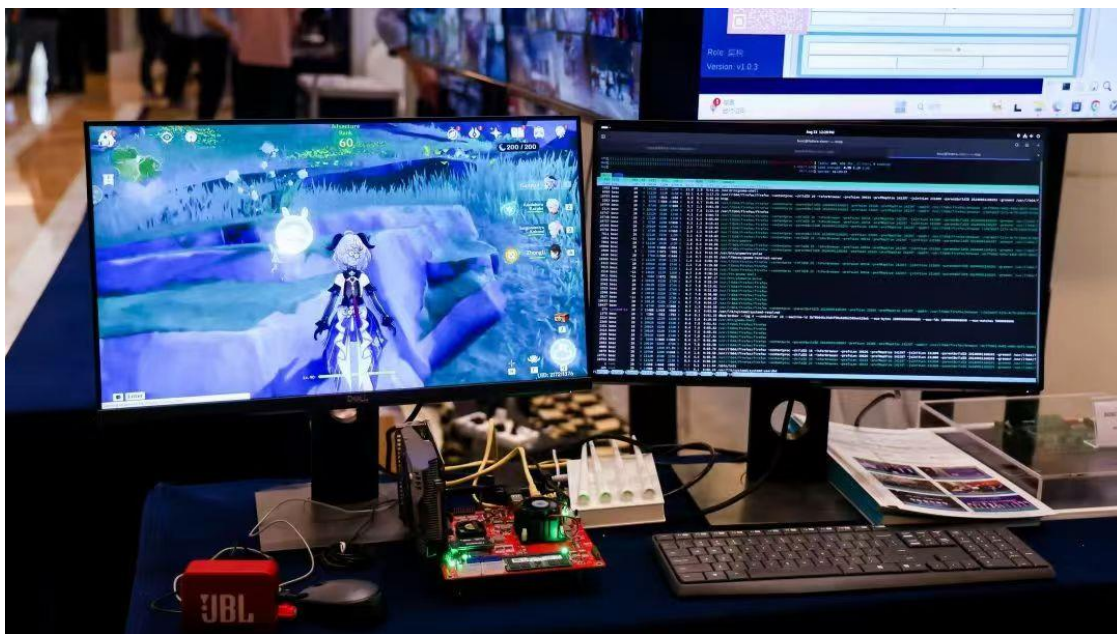
FPGA Prototype in Only Two Weeks



Source: Xinchon Technology

Real Chip of XiangShan Gen 2 Nanhu

- Test chip was back in October 2023
 - Successfully brought up Linux and working with PCIe device (GPU, Ethernet, USB..)



- Nanhu test board @ RVSC'24
 - Running Desktop Linux & Cloud Game*

The world's first laptop powered by a open-source RISC-V processor

ISCAS
中国科学院软件研究所
Institute of Software Chinese Academy of Sciences

Ruyi Book	
CPU	"XiangShan Nanhu" (RV64GCBK), up to 2.5GHz
Memory	8GB DDR5 4800MT/s
GPU	AMD RX 550
USB	2x USB3
Ethernet	2x 2.5Gbps Ethernet Port
Display	1x 14-inch LCD Display 1x HDMI, up to 4K
TouchPad	Support 9 kinds of gesture operation
Audio	Built-in high-quality speakers.
Dimensions	315*233*25mm

Powered by

XIANGSHAN
"XiangShan" high-performance open-source RISC-V processor

Ruyi Book

inchi 英趣智能
milkV

- Ruyi XiangShan Book
 - Design by ISCAS, Inchi, MilkV

* Genshin Impact · Cloud, only ~ 1 fps, just for fun

XiangShan Gen 3: Kunminghu

- **Target ARM Neoverse N2**
 - SPECCPU2006: 45@3GHz (15/GHz)
 - Vector/Hypervisor extension supported
- **A Joint Dev Team** (coordinated by BOSC)
 - About 10 institutions



Kunming Lake in Beijing



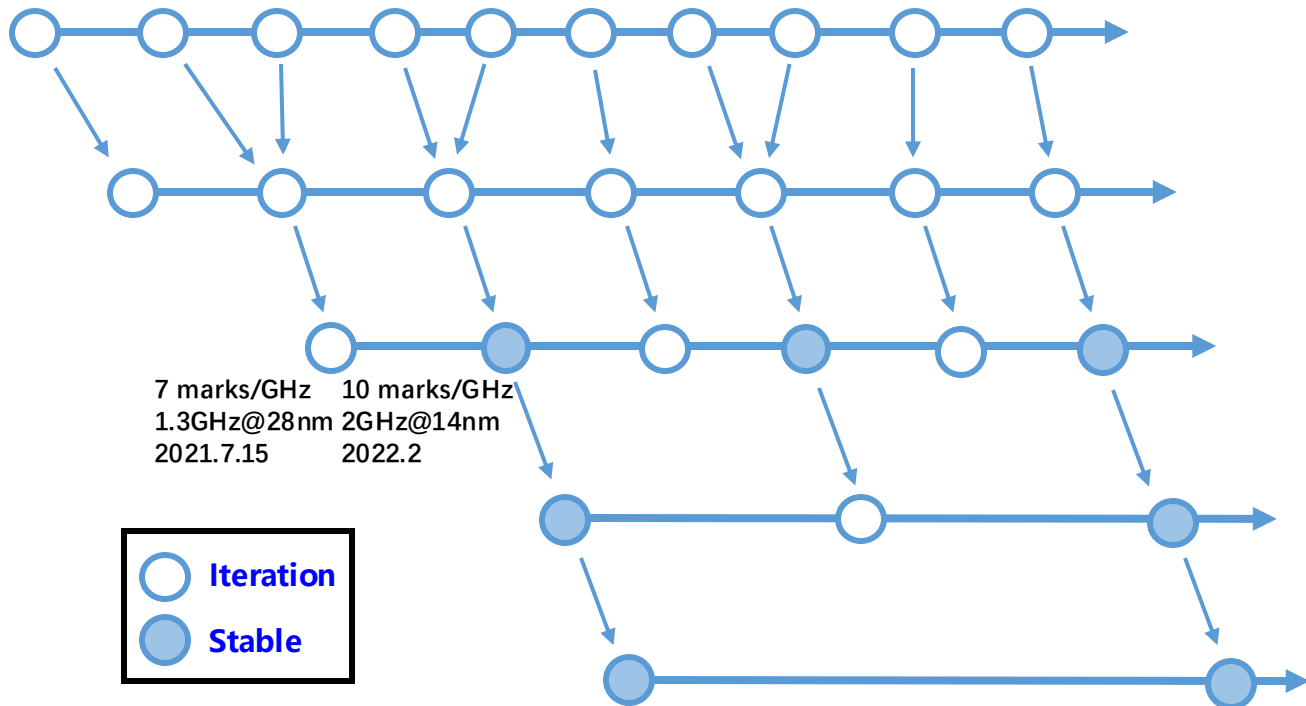
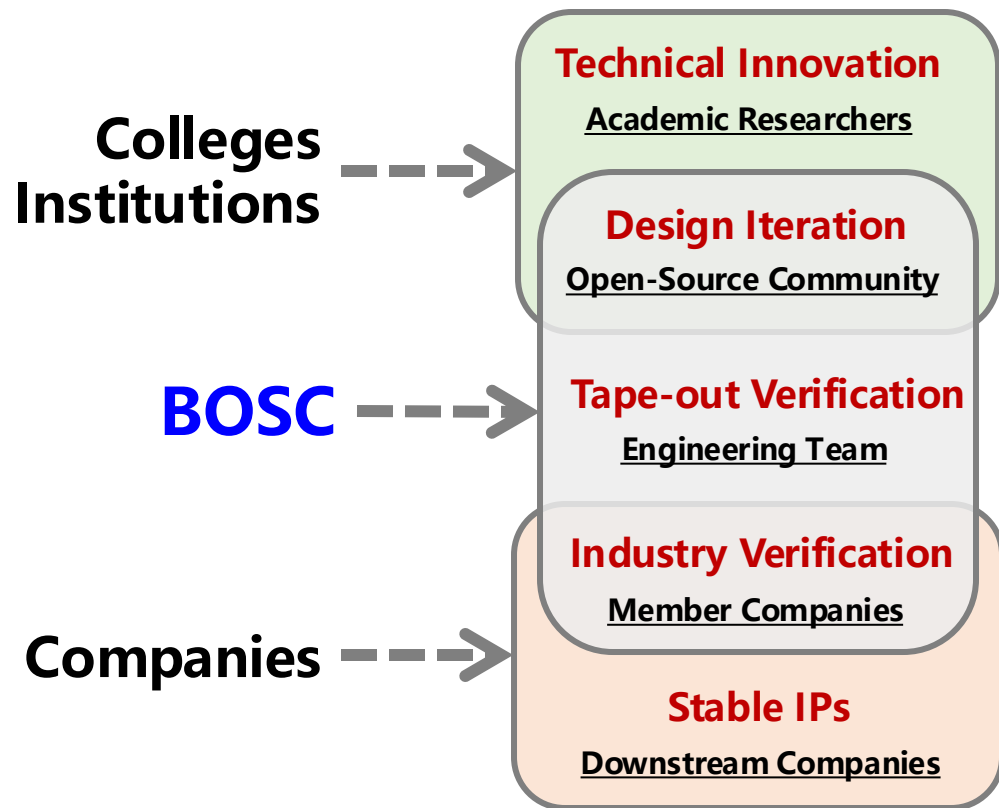


Collaborative Mechanism Based on Open-Source Innovation

- With high-level open-source hardware, we gather global innovative forces from various colleges and institutions, engaging in collaborative verification with the industry.



Tiered Rolling Open-Source Model





Highlights in XiangShan Gen 3 Kunminghu

- **Functional Enhancement**

- Support RISC-V **Vector/Hypervisor** extension
- Support **RVA23 profile**
- Support interconnection based on **CHI protocol**

- **Performance Exploration**

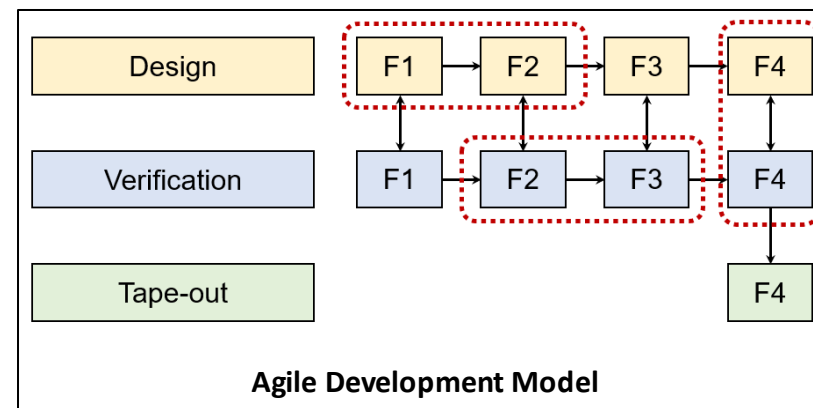
- Performance boost in frontend, backend, load-store unit and cache
- Performance model calibrated with RTL
- Workflow: **DSE on perf model => Impl. & fine tuning on RTL**

- **Functional Verification**

- **Hierarchical verification flow** spanning system/integration/unit level + FPGA prototyping
- Industrial-grade verification process

- **Physical Design**

- Experienced physical design team
- Simultaneous iteration of RTL coding based on timing evaluation

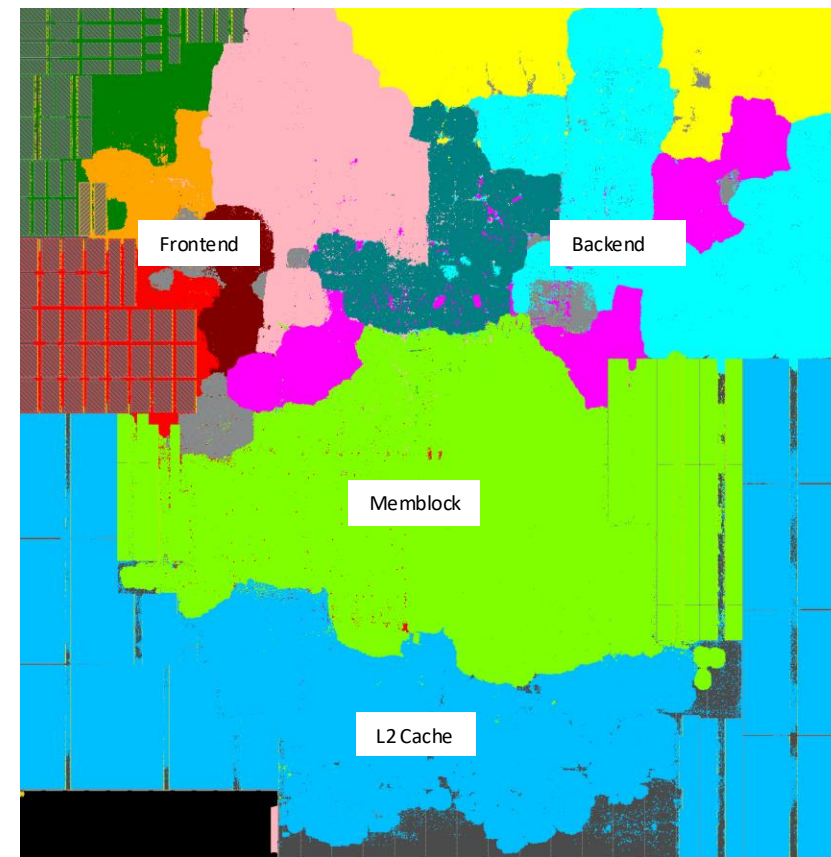




Performance Evaluation of Gen 3 Kunminghu

- Method: SPEC CPU checkpoints selected by Simpoint
 - Base: GCC 12 -O3, RV64GCB, jemalloc
 - 1MB L2 and 16MB L3
 - Simulated@3GHz with DRAMsim3 DDR4-3200

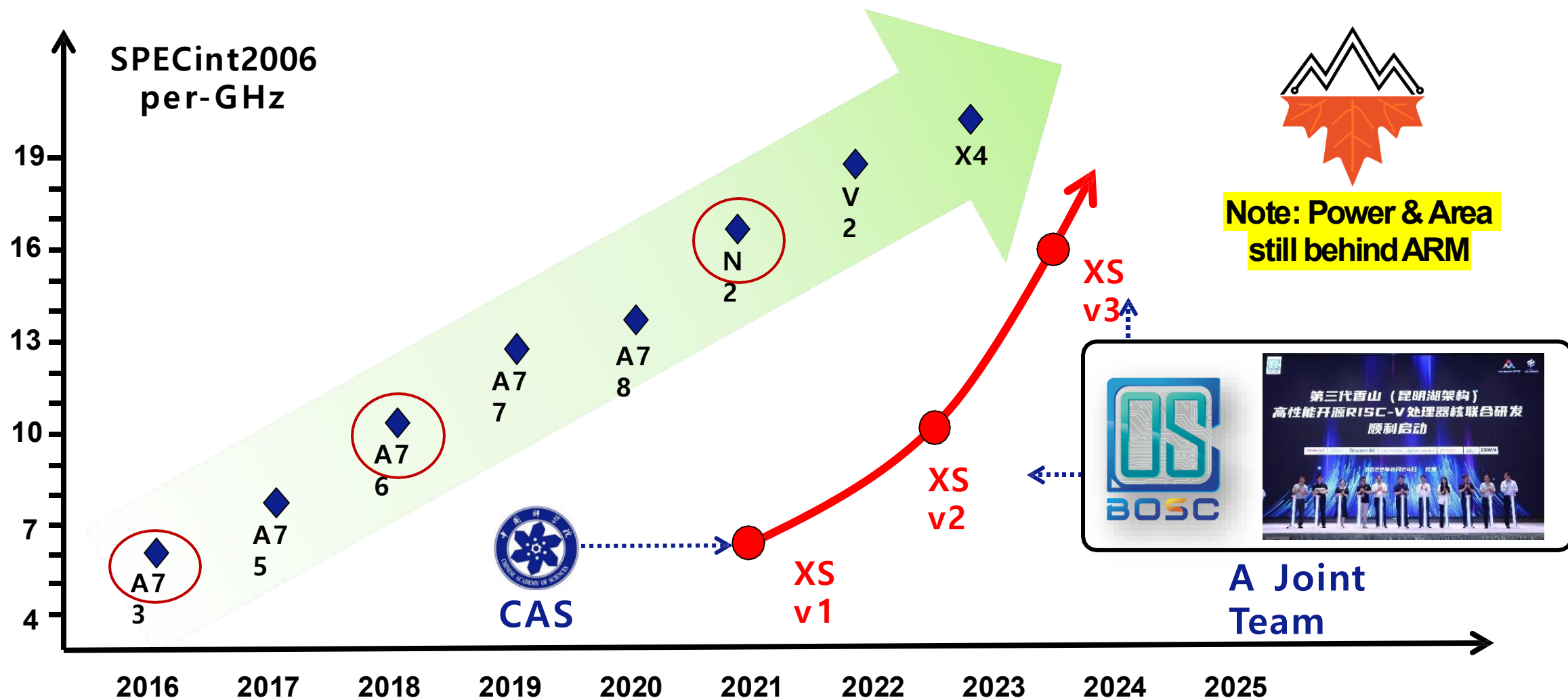
SPECint 2006 est.@ 3GHz		SPECfp 2006 est.@ 3GHz	
400.perlbench	35.88	410.bwaves	66.89
401.bzip2	25.50	416.gamess	40.89
403.gcc	46.72	433.milc	45.25
429.mcf	58.13	434.zeusmp	52.10
445.gobmk	30.26	435.gromacs	33.65
456.hmmer	41.60	436.cactusADM	46.16
458.sjeng	30.53	437.leslie3d	46.01
462.libquantum	122.50	444.namd	28.88
464.h264ref	56.57	447.dealII	73.43
471.omnetpp	39.37	450.soplex	51.99
473.astar	29.23	453.povray	53.44
483.xalancbmk	72.03	454.Calculix	16.38
GEOMEAN	44.15	459.GemsFDTD	37.18
		465.tonto	36.67
		470.lbm	91.24
		481.wrf	40.62
		482.sphinx3	48.57
		GEOMEAN	44.60



Floorplan of KMH V2R2 (single core)



XiangShan: Open-Source High Performance Processors



Prospect: "Dual Core" Roadmap



Based on **V3 (KMH)**

- **Big Core: Ultimate Performance (v.s. ARM N2)**

Target High-Throughput Advanced Computing Platform

Goal: become mainstream CPU for data centers and computational facilities



Based on **V2 (NH)**

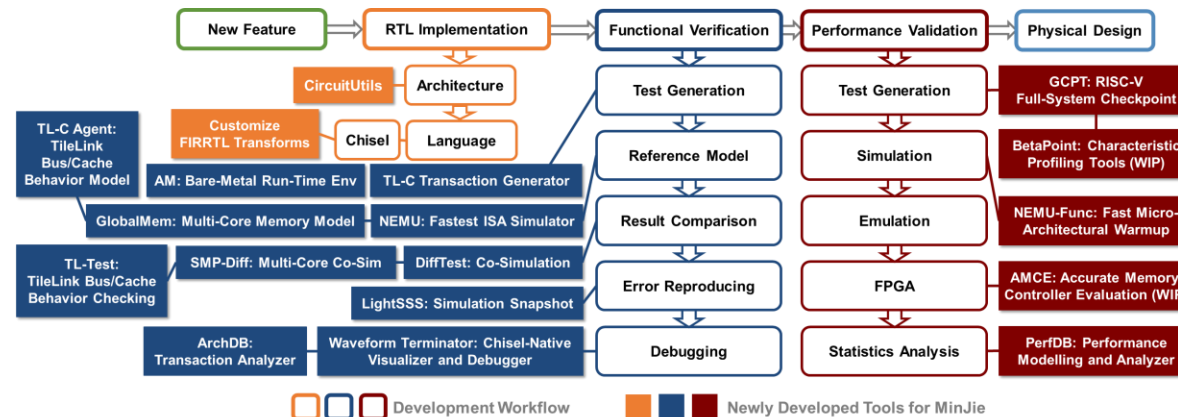
- **Mid Core: Balanced Perf & Efficiency (v.s. ARM A76)**

Target Mid-to-High-End General Industry Domain

Goal: support broad industrial spectrum including industrial control, automotive, communication, aviation and more

Part III

MinJie: Agile Development for High-Performance RISC-V Processors





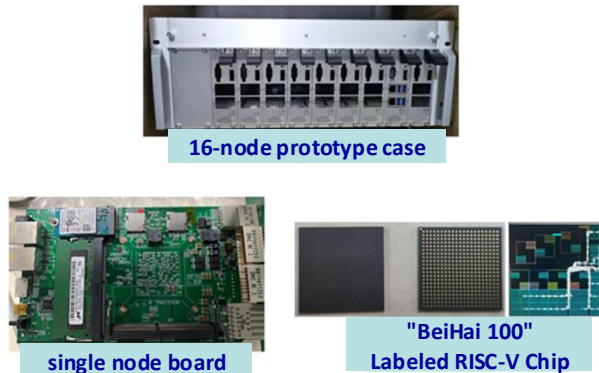
- **2020: 28-nm tape-out of an 8-core labeled RISC-V processor**

- **Task #2:** *Translate the Verilog codes into Chisel*
 - Evaluated on FPGA (xc7v2000tfg1716-1), Vivado 2017.01
- **Who:** *A junior student who never knew Chisel*

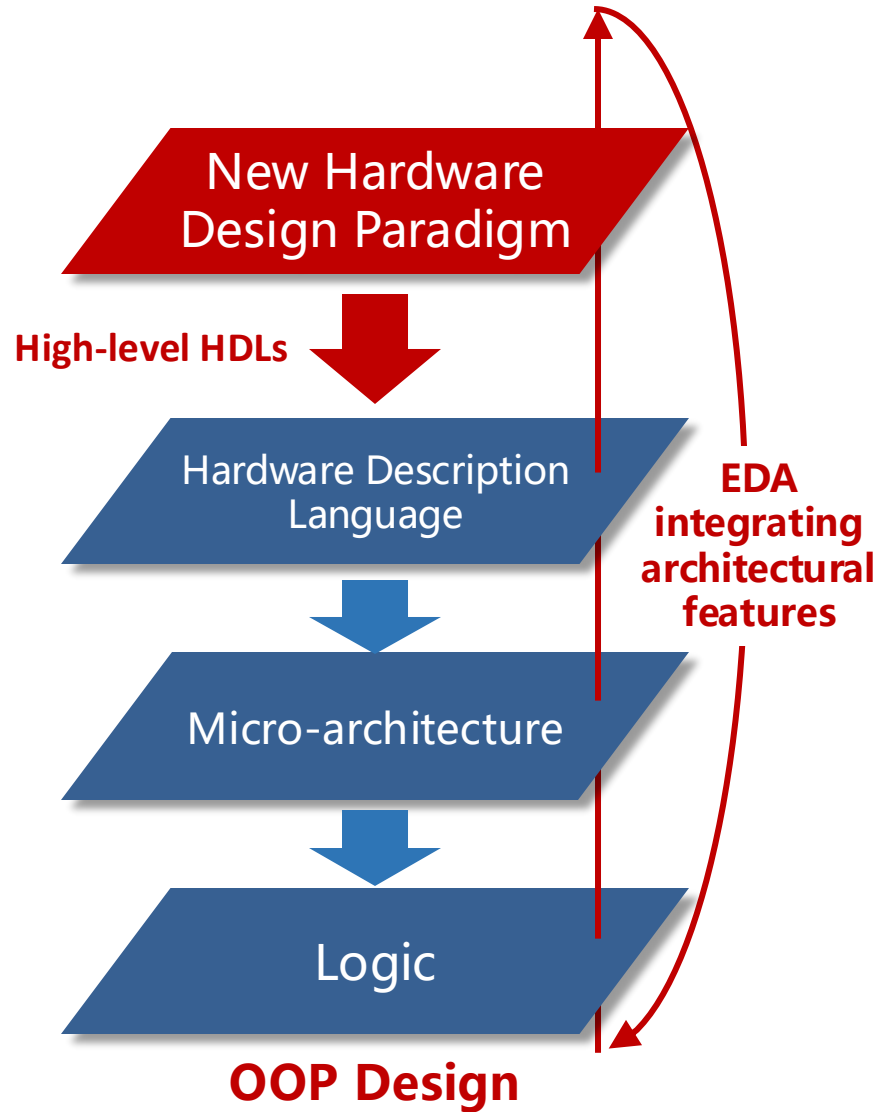
	Verilog	Chisel (direct translation)	Chisel-opt (adv. features & libs)
Freq./MHz	135.814	136.388 (+0.42%)	154.107 (+13.47%)
Power/W	0.770	0.749 (-2.73%)	0.749 (-2.73%)
LUT Logic	5676	6422 (+13.14%)	2594 (-54.30%)
LUT Storage	1796	1264 (-29.62%)	1492 (-16.93%)
FF	4266	3638 (-14.72%)	747 (-82.49%)
LOCS	618	470 (-23.95%)	155 (-74.92%)

- **2nd Round results:** *Chisel can achieve better PPA than verilog*

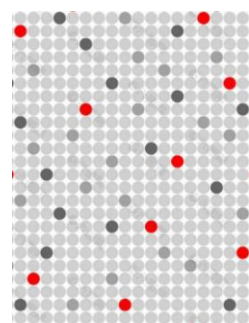
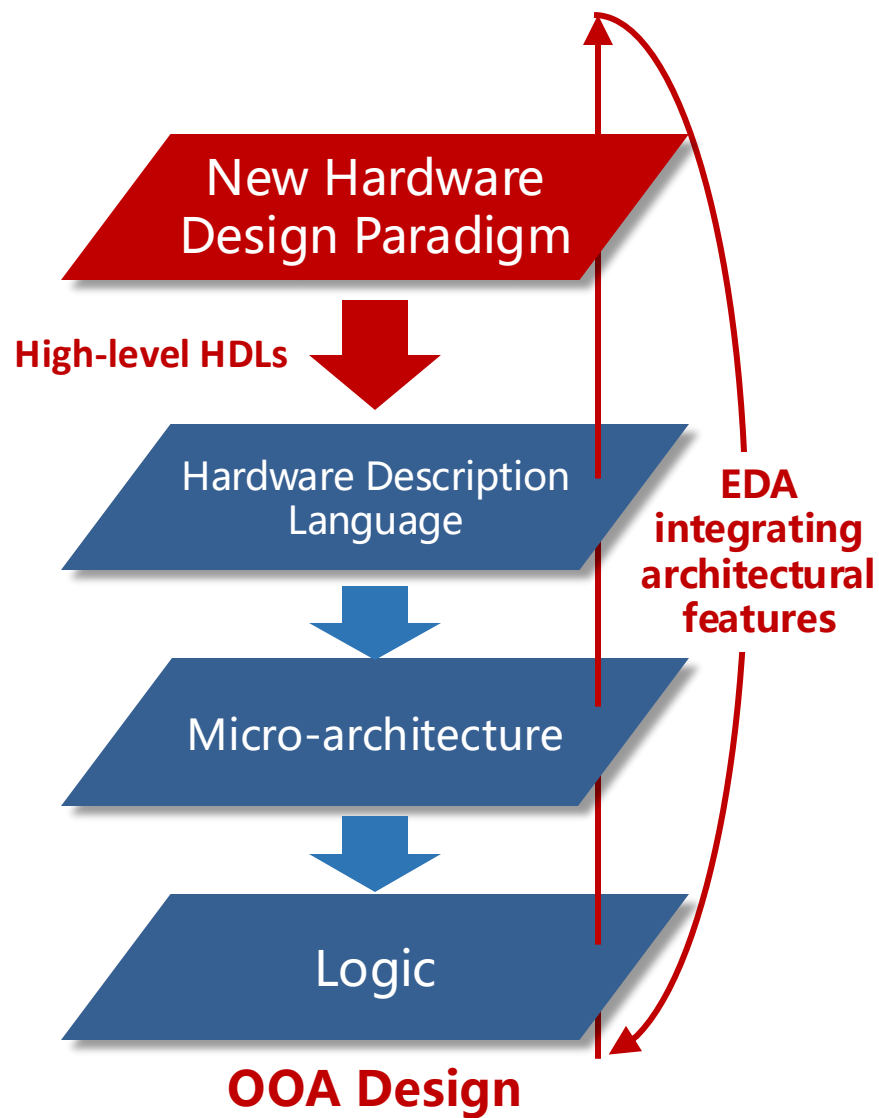
- RV64GC指令集
- 单发射顺序9级流水线
- 内置标签化冯诺依曼结构技术
- 8核/2MB L2 Cache
- ChipLink前端总线
- **1.2GHz@ 28nm**
- Wafer out/WB BGA封装
- 最大支持32GB DDR4内存
- 2*千兆以太网
- 1*PCIe3.0 RC x4



New HDL → New Design Paradigm



New HDL → New Design Paradigm

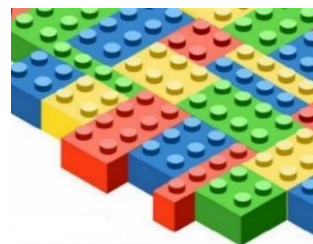


**Pixel
(gate-level)**

**Circuit-oriented
Processor Design**



Processor



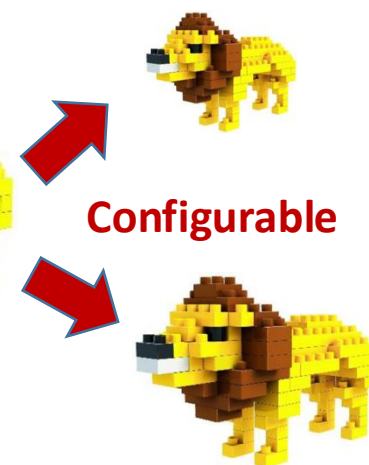
**Object
(Reusable Modules)**

**Object-oriented
Processor Design**



Processor

Configurable





What's Missing in Agile Hardware Design? Verification!

What's Missing in Agile Hardware Design? Verification!

Babak Falsafi, *Fellow, ACM, IEEE*

Parallel Systems Architecture Laboratory, Institute of Computer and Communication Sciences, School of Computer and Communication Sciences, Ecole polytechnique fédérale de Lausanne CH-1015 Lausanne

E-mail: babak.falsafi@epfl.ch

Agile hardware design is an approach to developing hardware systems that draws inspiration from the principles and practices of agile software development. It emphasizes collaboration, flexibility, iterative development, and quick adaptation to changing requirements. In agile hardware design, the focus is on delivering functional hardware systems in shorter development cycles while maintaining high-quality and customer satisfaction.

In particular, agile hardware design is of great interest in the open-source hardware community. Open-source hardware development —such as RISC-V— is at the forefront of initiatives to democratize hardware and drive innovation in chip design forward. Agile design is instrumental for the RISC-V community because it supports rapid iteration, accommodates the evolving RISC-V standard and the addition of custom extensions, improves community collaboration and time-to-market, and addresses the design challenges associated with complex architectural features.

Among significant innovations based on agile hardware design is the recently announced XIANGSHAN RISC-V core which is currently the highest performing RISC-V out-of-order microprocessor core with single-thread performance exceeding both existing RISC-V cores and a state-of-the-art ARM core, Cortex-A76. The creators of this platform have published their agile design methodology in a flagship computer architecture venue, MICRO, with a paper that has been selected through peer review to be among the best dozen papers in all of computer architecture in one year for publication in IEEE Micro Top Picks.

A key contributor to this breakthrough has been integrating hardware verification into the agile methodology. Hardware verification is crucial in designing digital platforms, as it ensures that semiconductor chips operate correctly and reliably according to the architecture specifications. Verification guarantees compliance with standards, and helps detect and rectify design errors, validate system-level functionality, optimize performance and power consumption, and enhance hardware reliability and safety. It plays a fundamental role in creating robust and dependable CPUs that meet the requirements of various applications and workloads.

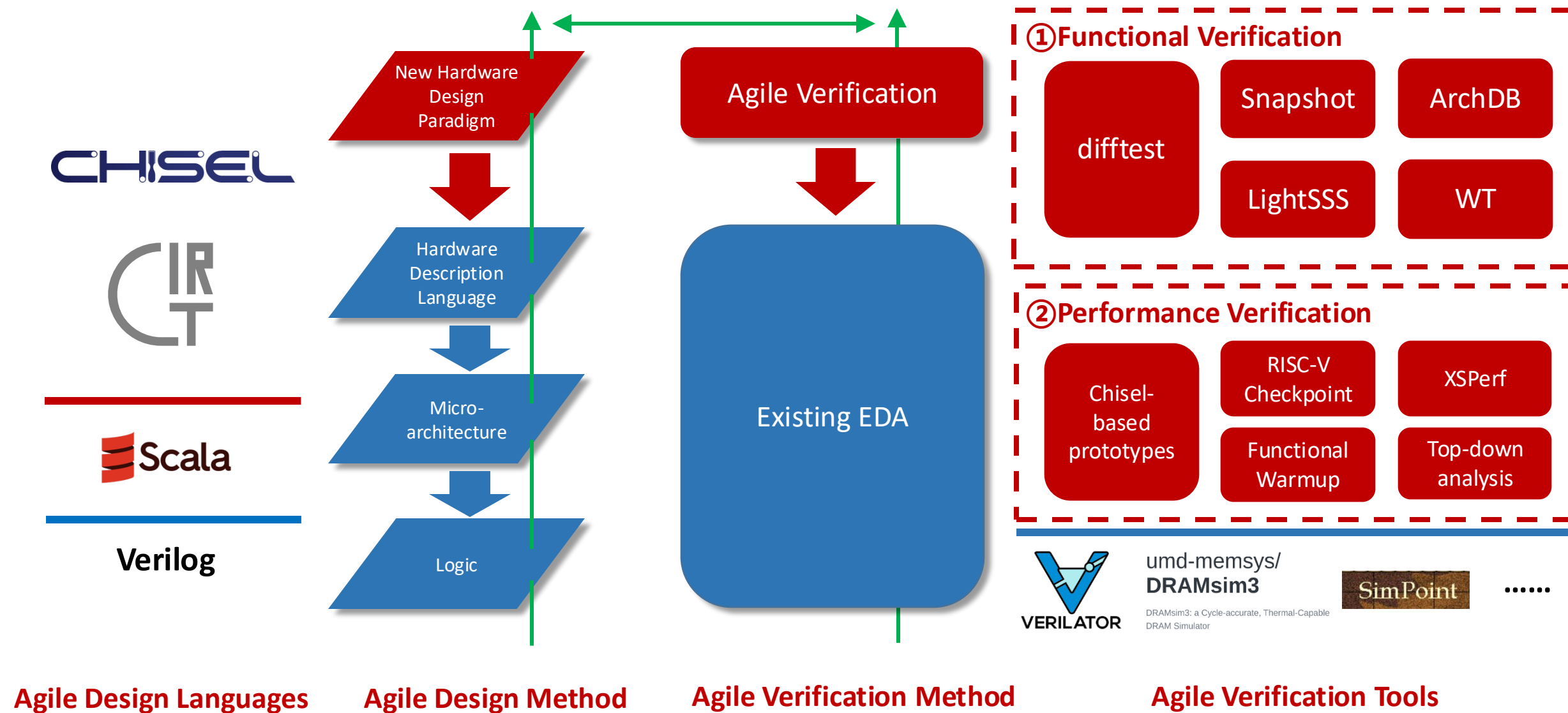


Babak Falsafi

Professor, EPFL

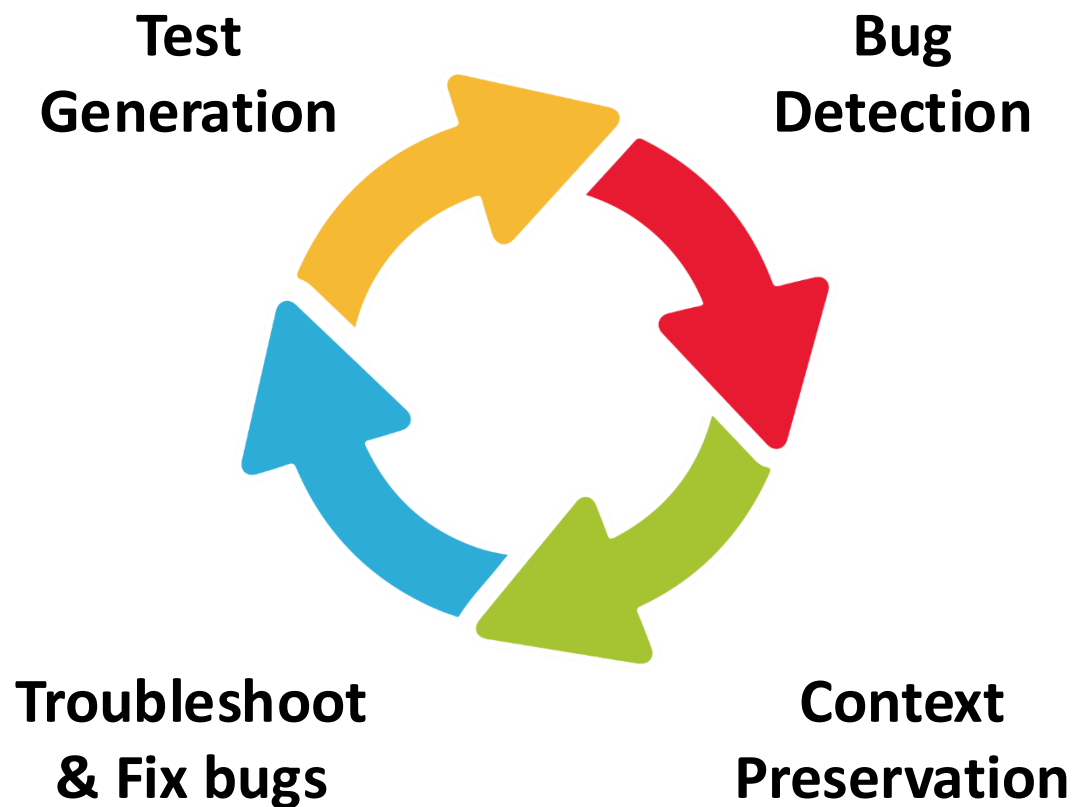
ACM/IEEE Fellow

Agile Verification is Challenging

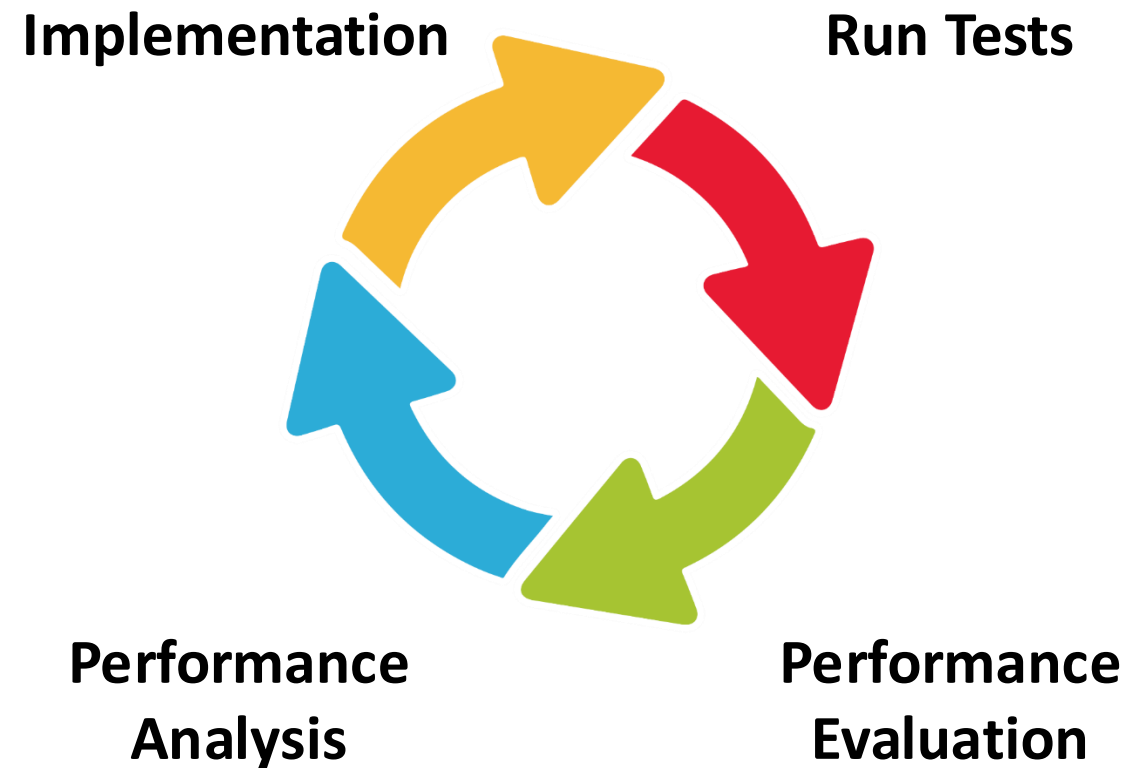


MinJie Loop

Functional Verification Loop



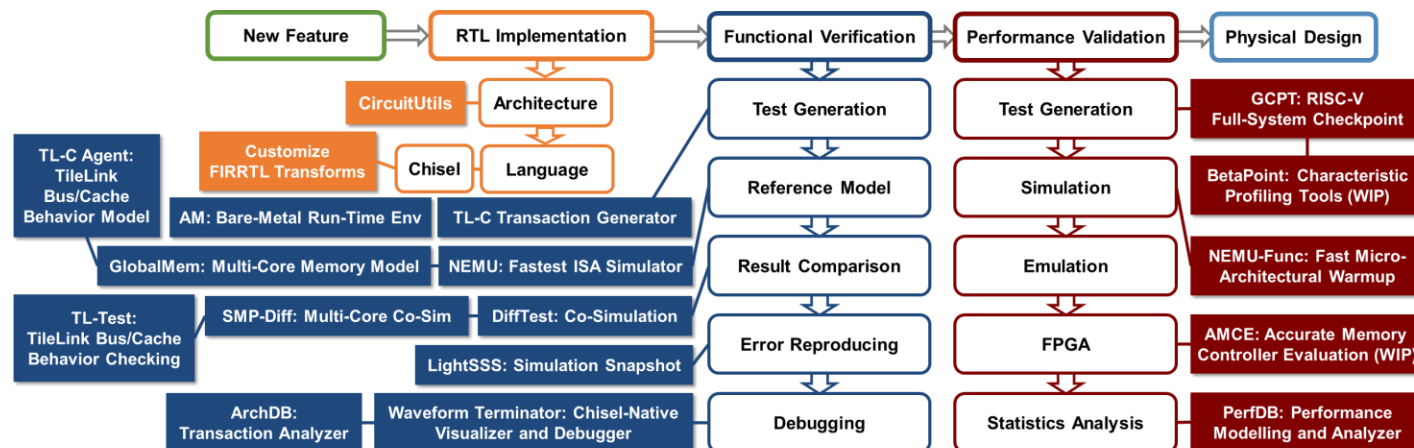
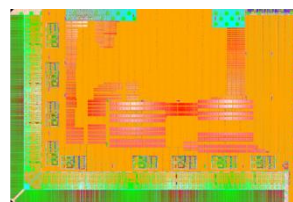
Performance Optimization Loop





Minjie: Open & Agile Verification Toolchain

- Infrastructure is the key outcome of the XiangShan Project
- Open source to benefit both academia and industry





Advanced Agile Chip Design Methodology

- New method and toolchain for agile chip design
- Developed more than 17 new tools to solve agile verification problems
- **MICRO 2022 → IEEE Micro Top Picks**

2022 55th IEEE/ACM International Symposium on Microarchitecture (MICRO)



Towards Developing High Performance RISC-V Processors Using Agile Methodology

Yinan Xu^{*†}, Zihao Yu^{*}, Dan Tang^{*‡}, Guokai Chen^{*†}, Lu Chen^{*†}, Lingrui Gou^{*†}, Yue Jin^{*†}, Qianruo Li^{*†}, Xin Li^{*†}, Zuojun Li^{*†}, Jiawei Lin^{*†}, Tong Liu^{*}, Zhigang Liu^{*}, Jiazhan Tan^{*}, Huaqiang Wang^{*†}, Huizhe Wang^{*†}, Kaifan Wang^{*†}, Chuanqi Zhang^{*†}, Fawang Zhang^{||}, Linjuan Zhang^{*†}, Zifei Zhang^{*†}, Yangyang Zhao^{*}, Yaoyang Zhou^{*†}, Yike Zhou^{*}, Jiangrui Zou^{||}, Ye Cai^{||}, Dandan Huan[¶], Zusong Li[¶], Jiye Zhao[¶], Zihao Chen[§], Wei He[§], Qiyuan Quan[§], Xingwu Liu^{**}, Sa Wang^{*†}, Kan Shi^{*}, Ninghui Sun^{*†} and Yungang Bao^{*†}

^{*}State Key Lab of Processors, Institute of Computing Technology, Chinese Academy of Sciences, China

[†]University of Chinese Academy of Sciences, China

[‡]Beijing Institute of Open Source Chip, China

[§]Peng Cheng Laboratory, China

[¶]Beijing VCore Technology Co., Ltd., China

^{||}Shenzhen University, China

^{**}Dalian University of Technology, China

THEME ARTICLE: TOP PICKS FROM THE 2022 COMPUTER ARCHITECTURE CONFERENCES

Toward Developing High-Performance RISC-V Processors Using Agile Methodology

Yinan Xu^{IC} and Zihao Yu^{IC}, State Key Lab of Processors, Institute of Computing Technology, Chinese Academy of Sciences, Beijing, 100190, China

Dan Tang^{IC}, Beijing Institute of Open Source Chip, Beijing, 100080, China

Ye Cai^{IC}, Shenzhen University, Shenzhen, 518060, China

Dandan Huan^{IC}, Beijing VCore Technology, Beijing, 100190, China

Wei He^{IC}, Peng Cheng Laboratory, Shenzhen, 518060, China

Ninghui Sun^{IC} and Yungang Bao^{IC}, State Key Lab of Processors, Institute of Computing Technology, Chinese Academy of Sciences, Beijing, 100190, China



XiangShan achieves L2.5

L1: OPEN ISA



L2: OPEN Design/Implementation



L3: OPEN Framework/Tools



3

Open Framework/Tools

microarch.

coding

EDA Tools

ISA Spec.

The RISC-V Instruction Set Manual
Volume I: User-Level ISA
Document Version 2.2

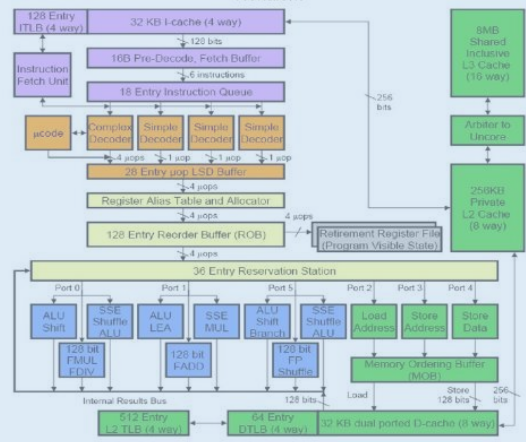
Editors: Andrew Waterman^a, Kristof Arzoo^{a,b}

^aCS Division, EECS Department, University of California, Berkeley
arwater@eecs.berkeley.edu, kristof@eecs.berkeley.edu
May 7, 2017

1

Open ISA

Docs



2

Open Design/Implt

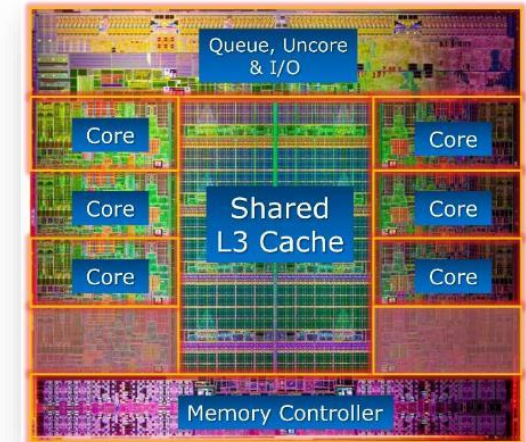
RTL

```
component DebugCoreTop is
  port (
    -- Trigger and Data
    cu_Clk      : in    std_logic_vector(2 downto 0) := (others => '0');
    cu0_Trig    : in    t_trig_0 := (others => (others => '0'));
    cu1_Trig    : in    t_trig_1 := (others => (others => '0'));
    cu2_Trig    : in    t_trig_2 := (others => (others => '0'));
    cu0_Data    : in    t_data_0 := (others => (others => '0'));
    cu1_Data    : in    t_data_1 := (others => (others => '0'));
    cu2_Data    : in    t_data_2 := (others => (others => '0'));

    -- Downstream I2C
    SCL         : in    std_logic := '0';
    SDA         : inout std_logic := '0';

    -- Upstream
    gt_RefClk_p : in    std_logic := '0';
    gt_RefClk_n : in    std_logic := '0';
    gt_RX_p     : in    std_logic_vector(2 downto 0) := (others => '0');
    gt_RX_n     : in    std_logic_vector(2 downto 0) := (others => '0');
    gt_TX_p     : out   std_logic_vector(2 downto 0);
    gt_TX_n     : out   std_logic_vector(2 downto 0);
  );
end component;
```

Layout



Part IV

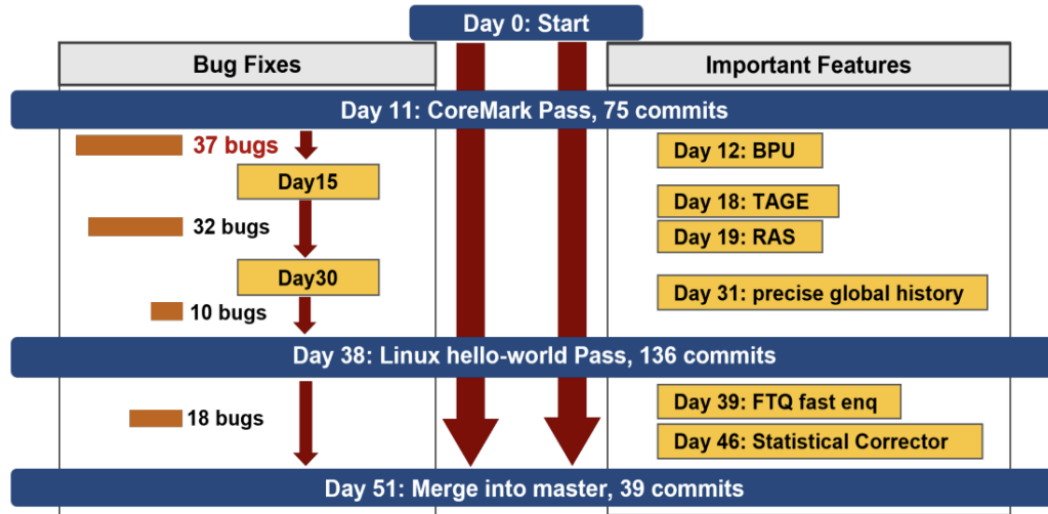
An Effective Infrastructure for Research





An Effective Infrastructure for Research

① Micro-architecture Optimization



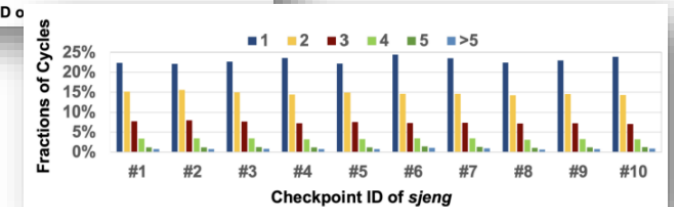
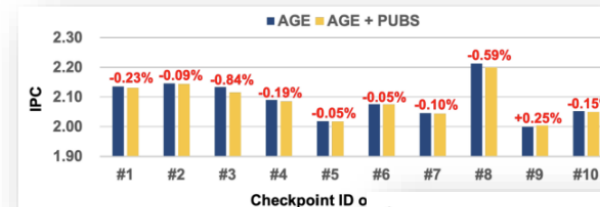
- 3 graduate students
- 11 days for a functionally correct prototype
- 37 bugs in 5 days
- 38 days to boot Linux with BPU
- 51 days for the overall frontend architecture

② Paper Reproduction

2018 51st Annual IEEE/ACM International Symposium on Microarchitecture

Performance Improvement by Prioritizing the Issue of the Instructions in Unconfident Branch Slices

One third-year PhD student, 200 minutes on XiangShan



An Effective Infrastructure for Research

- **Topic: Computer Architecture**

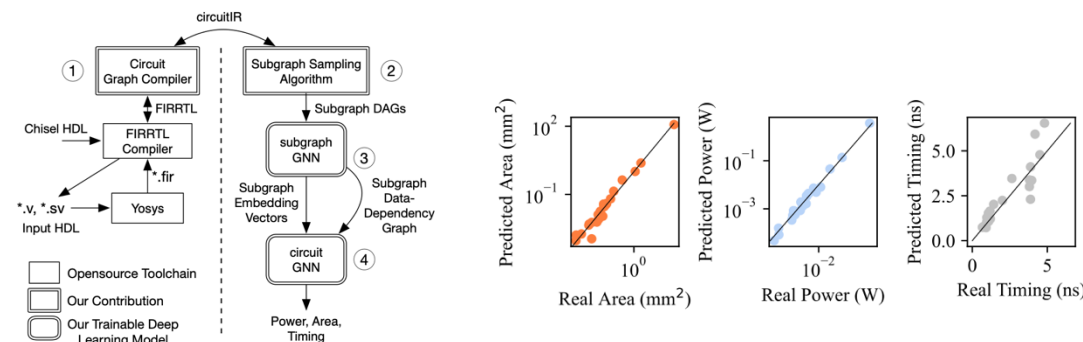
- XiangShan: a realistic out-of-order RISC-V implementation with industry-competitive performance and an active open-source community
- MinJie provides the toolchains
- *Microarchitecture, accelerators, novel architectures, profiling, systems, benchmarking, security, compilers, ...*



Imprecise Store Exceptions, ISCA'23 (EPFL)

- **Topic: Agile Chip Development**

- XiangShan is a progressive, configurable, complicated, challenging benchmark
- MinJie provides a good startpoint
- *HDLs, verification, performance, power, area, prototyping, DFT, synthesis, placement, routing, ECO, ...*



SNS v2, MICRO'23 (Duke University)

Summary

- **Era of open-source chip is coming**
XiangShan fills the gap in high performance open source processors.
- **Three generation, dual-core roadmap**
XiangShan meet the needs from both academia and industry.
- **An effective platform for research on microarchitecture and agile hardware design.**



Thanks!